OSINT & GOOGLE DORKING

1. Introduction to OSINT

- OSINT stands for Open-Source Intelligence.
- It refers to collecting and analyzing information from publicly available sources.
- The goal of OSINT is to turn openly available data into useful intelligence.
- It plays a vital role in cybersecurity, threat intelligence, fraud investigation, risk assessment and more.

Key Idea

If it is public, it can be found. If it is found, it can be used (ethically).

OSINT = DATA → INFORMATION → INTELLIGENCE

2. Why OSINT is Important?

- Helps in understanding threats and attackers.
- Identifies exposed or leaked information.
- Supports security assessments and audits.
- Useful for investigation and incident response.
- Helps organizations make informed decisions.

Benefits

- Cost effective
- Legal and ethical (when done properly)
- Wide range of sources
- Real-time information

3. Types of OSINT Sources

Public Web	Social Media	Public Records	Technical Sources	Media & Archives
Websites, Blogs, News sites, Forums, Online Documents.	Platforms like Facebook, Twitter (X), LinkedIn, Instagram, etc.	Govt, records, Court documents, Business registries, Voter lists, etc.	Domain info (WHOIS), DNS records, IP info, Shodan, Censys, Certificates.	News archives, Wayback Machine, YouTube, Podcasts, Press releases.

4. OSINT Lifecycle

Planning → **Collection** → **Processing** → **Analysis** → **Reporting**

Planning	Collection	Processing	Analysis	Reporting
Define objective, scope, target, and legal boundaries.	Gather data from various public sources.	Organize, filter and clean the collected data.	Analyze the data to find patterns, connections and insights.	Document findings in a clear and structured format.

5. OSINT Collection Techniques

- Direct Search – Using search engines
- Social Media Profiling – Analyzing user info
- Website Footprinting – Extracting details about websites
- Metadata Analysis – Checking hidden information in files
- Public Records Search – Using government and public databases
- Archive Search – Using tools like Wayback Machine
- Domain & IP Recon – WHOIS, DNS, IP lookup

Note

Always stay within legal and ethical boundaries. Collect only what is publicly available. Do not hack, exploit or access private systems.

6. Google Dorking – Introduction

- Google Dorking is the practice of using advanced search operators to find specific information on the internet.
- It helps in finding hidden pages, sensitive files, login portals, leaked data, misconfigurations and more.
- Dorking is not hacking. It uses Google's search engine in a smart way.

Important

Use Google Dorking only on systems and domains you are authorized to test.

2. WHY OSINT IS IMPORTANT?

- Helps in understanding threats and attackers.
- Identifies exposed or leaked information before it is exploited.
- Supports security assessments and audits.
- Useful for investigation and incident response.
- Helps organizations make informed decisions based on real-world data.
- Reduces risk by identifying vulnerabilities early.
- Cost effective – uses publicly available information.
- Improves situational awareness and cyber threat intelligence.

BENEFITS OF OSINT

- Cost effective (uses free/public data)
- Legal and ethical (when done properly)
- Wide range of sources
- Real-time information
- Helps in proactive security
- Supports threat hunting
- Aids in fraud prevention
- Enhances decision making

OSINT VS CLOSED-SOURCE INTELLIGENCE

OSINT (Open-Source)	Closed-Source (Proprietary)
• Publicly available information	• Restricted / paid information
• Low cost or free	• High cost
• Easy to access	• Harder to access
• May not always be 100% accurate	• Usually more accurate
• Larger in volume	• Smaller in volume
• Legal to use	• Access requires authorization

REMEMBER

Good OSINT leads to better intelligence, Better intelligence leads to better decisions, and better decisions lead to better security.

☆

EXAMPLES OF OSINT IN REAL WORLD

→ A security analyst searches LinkedIn, job portals, and company websites to know the technologies used by a company.

→ An investigator uses public court records and news articles to gather case information.

→ A researcher checks domain details, DNS records and website data to find potential vulnerabilities.

→ A fraud analyst reviews social media and public complaints to identify suspicious activities.

KEY TAKEAWAY

OSINT is the foundation of modern cybersecurity and investigations. It turns publicly available information into actionable intelligence.

NOTE

Always use OSINT ethically, legally and responsibly. Respect privacy and obey all applicable laws.

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3. TYPES OF OSINT SOURCES

OSINT data comes from a wide variety of public and openly available sources.

1. PUBLIC WEB

- Websites
- Blogs
- News sites
- Forums
- Online documents

Examples:

- Company websites
- Press releases
- Public reports
- Whitepapers

2. SOCIAL MEDIA

- Platforms where users share information
- Useful for user profiling, monitoring & trend analysis

Examples:

- Facebook
- Twitter (X)
- LinkedIn
- Instagram
- YouTube
- Reddit

3. PUBLIC RECORDS

- Government and public databases
- Records legally accessible to citizens
- Valuable for verification and background research

Examples:

- Court documents
- Business registries
- Property records
- Voter lists
- Patents & licenses

4. TECHNICAL SOURCES

- Technical information about domains, networks & systems
- Helps in infrastructure reconnaissance

Examples:

- WHOIS records
- DNS records
- IP information
- Certificates (SSL/TLS)
- Shodan, Censys

5. MEDIA & ARCHIVES

- Historical and archived information
- Useful for trend analysis and investigations

Examples:

- News archives
- Wayback Machine
- Podcasts
- Press releases
- Online libraries

OTHER IMPORTANT OSINT SOURCES

→ **Search Engines** : Google, Bing, DuckDuckGo, Yandex (Use advanced operators for better results)

→ **Academic & Research** : Journals, conference papers, universities, research repositories (e.g., ResearchGate)

→ **Job Portals** : LinkedIn Jobs, Indeed, Glassdoor – reveal company structure & technologies.

→ **Open Data Portals** : Data.gov, data.gov.in – government and public datasets.

→ **Cloud Storage & Code Repos** : GitHub, GitLab, Pastebin, Gist – may contain exposed data or code.

→ **Maps & Geolocation** : Google Maps, OpenStreetMap – useful for location intelligence.

→ **Email & Communication** : Public email formats, contact forms, mailing lists.

→ **Breach Databases** : HaveIBeenPwned, LeakCheck – check leaked credentials.

KEY POINTS

✓ No single source gives complete information.

✓ Combining multiple sources gives better intelligence.

✓ Always evaluate the reliability of the source.

✓ Information should be verified before use.

✓ Respect privacy and legal boundaries.

REMEMBER

OSINT is not about collecting as much data as possible, but about collecting the right data from the right sources.

☆ **NOTE** : Always use publicly available information and respect privacy laws. Do not access private or restricted systems.

4. OSINT LIFECYCLE

- OSINT follows a structured lifecycle to convert raw data into useful intelligence.

GOAL

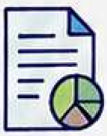
To produce accurate, reliable and actionable intelligence using publicly available information.

1. PLANNING →
- Define the objective, scope, and goal.
- Identify key questions to answer.
- Determine legal boundaries and rules.

2. COLLECTION →
- Gather data from various open sources.
- Use search engines, websites, databases, social media, etc.
- Record the source of each data.

3. PROCESSING →
- Organize and filter the collected data.
- Remove duplicates and irrelevant data.
- Extract useful information.
- Normalize data for analysis.

4. ANALYSIS →
- Analyze the information to find patterns, relationships and connections.
- Correlate data from multiple sources.
- Assess the credibility and reliability.



5. REPORTING →
- Document the findings in a clear format.
- Include sources and evidence.
- Present actionable intelligence.
- Update reports when new data is available.

KEY POINTS

✓ Plan before you collect.

✓ Collect widely but within legal limits.

✓ Process carefully to remove noise.

✓ Analyze critically, not emotionally.

✓ Report clearly and with evidence.

LIFECYCLE EXPLAINED

1 PLANNING	2 COLLECTION	3 PROCESSING	4 ANALYSIS	5 REPORTING
You decide what you need to find and how you will find it.	You gather as much relevant data as possible.	You clean and organize the data for better understanding.	You find patterns, connections and hidden insights.	You prepare the final report for decision making.
Example: Find emails of company employees.	Example: Using Google, LinkedIn, company website, etc.	Example: Filtering only valid email addresses.	Example: Mapping employee structure.	Example: Report with findings and source links.

REMEMBER

Good OSINT is not about finding everything, but about finding the right information from the right sources at the right time.

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NOTE

- Always follow legal and ethical guidelines.
- Do not access private or restricted systems.
- Respect privacy and data protection laws.
- Verify information from multiple sources.

5. OSINT COLLECTION TECHNIQUES

- Collection is the most important phase in OSINT. The quality of intelligence depends on how good your data collection is.

TECHNIQUE	DESCRIPTION	HOW TO USE / EXAMPLES
1. Search Engine Searching	Use search engines to find information from websites, documents, images, news etc.	→ Use Google, Bing with advanced operators (Google Dorking) → Search for documents, pages, directories, leaks etc.
2. Social Media Profiling	Collect information from social media platforms about people, companies or organizations.	→ Check public profiles, posts, comments, likes, shares. → Look for patterns, connections, location, interests. → Platforms: Facebook, Twitter (X), LinkedIn, Instagram, YouTube, Reddit.
3. Website Footprinting	Extract useful information from the target website or domain.	→ Check "About Us", "Contact Us", terms, privacy policy. → Look for hidden pages, directories, subdomains, metadata. → Use tools like BuiltWith, Wappalyzer.
4. Document Analysis	Analyze public documents to extract useful information.	→ PDFs, DOCX, PPT, XLS files may contain metadata. → Use tools like ExifTool, PDF Metadata, Office Metadata. → Look for author name, email, location, software used.
5. Public Records Search	Search public databases and government records for valid information.	→ Court records, business registrations, voter lists, property records, patents, licenses. → Use official government websites and portals.
6. Email & Username Hunting	Find email addresses or usernames using public sources.	→ Check websites, contact pages, social media, forums. → Use tools like Hunter.io, Namechk, theHarvester. → Search for patterns (e.g., name@domain.com).
7. Archive & Historical Search	Find old or deleted information using archives and historical records.	→ Use Wayback Machine, Archive.today, Cached pages. → Check deleted posts, old web pages, domain history. → Useful for trend analysis & investigations.

TIPS

- ✓ Be patient and thorough.
- ✓ Collect more, analyze later.
- ✓ Keep data organized.
- ✓ Always verify before trusting the data.

☆

NOTE

Never use hacking or intrusive methods. OSINT is about using publicly available information only.

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BEST PRACTICES FOR COLLECTION

- Define what information you need before starting.
- Use multiple sources to confirm the same information.
- Collect data ethically and legally.
- Document the source of every information collected.
- Maintain proper notes and evidence for reporting.

REMEMBER

Wide collection + Proper verification
= High quality intelligence

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COMMON DATA COLLECTION TOOLS (EXAMPLES)

- Google / Bing – General information search
- Google Dorking – Advanced search using operators
- BuiltWith – Technology used by websites
- Wappalyzer – Identify web technologies
- Hunter.io – Find email addresses
- theHarvester – Email, subdomain & host discovery
- Wayback Machine – View historical versions of websites
- ExifTool – Extract metadata from files
- DomainTools – Domain & IP related information
- Shodan – Search internet connected devices
- HaveIBeenPwned – Check email or domain breaches

☆ Always collect data with a purpose, follow legal boundaries and respect privacy. Ethical collection makes OSINT powerful and trustworthy. ☺

6. Introduction to Google Dorking

- Google Dorking is the practice of using advanced search operators and filters in Google to find specific information that is not easily found through normal searches.
- It helps in finding hidden or unindexed content, sensitive files, login pages, databases, and misconfigured websites.
- Dorks do not hack anything; they only use publicly available information.
- Google is the world's most powerful search engine. Using the right dorks = better results.

Goal

To find specific, relevant and actionable information using Google search operators.

Remember

Google dorking is for Defensive Security, Research, and Ethical Investigations only.

How Google Dorking Works?

You enter a query with operators → Google applies filters based on operators → It returns results matching those filters → You analyze and extract useful information

Common Google Dorking Operators

Operator	Description	Example	What It Does
site:	Search results from a specific website	site:example.com	Shows results only from example.com
intitle:	Search pages with a word in the title	intitle:"admin login"	Finds pages having "admin login" in the title
inurl:	Search pages with a word in the URL	inurl:login	Finds pages having "login" in the URL
filetype:	Search for specific file types	filetype:pdf	Finds only PDF files
" "	Search for exact phrase	"confidential report"	Finds pages with exact phrase
–	Exclude a word	hacking –tutorial	Shows results for "hacking" but excludes "tutorial"
OR	Search one term or another	admin OR login	Results with either admin or login
*	Wildcard (unknown word)	best * tools	Best [anything] tools (best osint tools, etc.)
..	Range of numbers	cisco ports 20..25	Shows results containing numbers from 20 to 25

Tips

- ✓ Combine multiple operators for better results.
- ✓ Keep queries simple and specific.
- ✓ Use quotes for exact matches.
- ✓ Verify information from multiple sources.
- ✓ Don't rely on a single search.

Note

- Dorking cannot access restricted or private data.
- Results depend on what is publicly available.
- Ethical use is important – respect privacy and follow laws.
- Always use dorks for learning, research, and defensive purposes only. ☺

7. Google Dorking – Advanced Operators

- Advanced operators help in narrowing down results and finding highly specific information.

OPERATOR	DESCRIPTION	EXAMPLE	WHAT IT DOES
intext:	Search for a word or phrase anywhere in the content.	intext:confidential report	Finds pages containing the words "confidential report" in the content.
allintitle:	All the words must be in the title.	allintitle:login portal	Finds pages with both "login" and "portal" in the title.
allinurl:	All the words must be in the URL.	allinurl:admin login	Finds pages with both words in the URL.
inanchor:	Search for text in links (anchor text).	inanchor:download manual	Finds pages that have the text in links pointing to them.
related:	Find websites related to a domain.	related:example.com	Shows websites similar to the given domain.
cache:	Show cached version of a web page.	cache:example.com/page.html	Displays the last cached copy of that page.
define:	Get definition of a word.	define:phishing	Shows dictionary definition.
info:	Get information about a specific page.	info:example.com	Shows cached info, similar pages and related details.
before: & after:	Find pages within a specific time range.	site:example.com report before:2023	Finds results before a specific date.
AROUND(X)	Finds two words within X words of each other.	data AROUND(5) breach	Finds pages where "data" and "breach" appear within 5 words.

Remember

Use quotes (" ") for exact phrases. Combine operators for more precise results.

Note

Google may ignore some operators if used incorrectly. Always keep your search simple, logical and targeted.

Combining Operators – Examples

- site:example.com filetype:pdf "annual report" → Find PDF annual reports from a specific website.
- intitle:"index of" inurl:/backup/ → Find index of backup directories.
- intext:"confidential" filetype:docx → Find Word documents containing confidential.
- site:gov.in inurl:tender filetype:pdf → Find government tender PDFs.
- intitle:login OR intitle:sign in → Find pages with login OR sign in in title.

Best Practices

✓ Start broad, then narrow down.
✓ Use multiple operators together.
✓ Verify information from multiple sources.
✓ Do not rely only on Google.
✓ Keep queries specific and relevant.

Example – Practical Dorks (Safe & Legal)

★ filetype:pdf "public policy" site:gov.in
★ inurl:research filetype:pdf site:edu
★ intitle:"privacy policy" site:example.com
★ intext:contact us intitle:directory
★ site:*.gov "press release" after:2023

★ Google Dorking is a powerful research technique – When used ethically and legally, it can uncover valuable and actionable intelligence.

8. OSINT Tools – Categories & Examples

- OSINT tools help automate searching, collect data, analyze, and visualize information from multiple sources.

CATEGORY	DESCRIPTION	POPULAR TOOLS & EXAMPLES	WHAT THEY DO
Search Engines	Advanced search and dorking capabilities	Google, Bing, DuckDuckGo, Yandex, Shodan	Find websites, pages, documents, info using advanced queries
Social Media Intelligence	Information from social platforms	Maltego, Sherlock, TweetDeck, Hootsuite, Twint	Find profiles, posts, interactions, connections
Data & Documents Search	Find public documents and data repositories	Google Dorking, WorldCat, Open Government Data, Scribd	Locate PDFs, reports, spreadsheets, legal docs
Mapping & Location Intelligence	Geospatial data and location information	Google Maps, OpenStreetMap, Geolocated Photos, Wikimapia	Find locations, places, coordinates, satellite imagery
Network & IP Intelligence	Domain, IP, DNS and network information	WHOIS, DNS Lookup, IP2Location, Shodan, Censys	Find domain info, IP details, DNS records
</> Code & Developer Tools	Code search and developer data	GitHub, GitLab, Pastebin, Stack Overflow, SourceForge	Find code, leaks, repositories, credentials
Open Data & APIs	Government and organization datasets	data.gov, OpenStreetMap, World Bank Data, Kaggle	Access datasets for analysis and research
Media & Archives	News archives and historical data	Wayback Machine, Archive.today, Google News Archive	Find historical versions and old media
Communication Intelligence	Emails and public communication data	HaveIBeenPwned, Hunter.io, Email Checker	Find email leaks, verify email existence

Popular OSINT Tools (General List)

- Maltego – Link analysis and entity mapping
- Shodan – Search devices and internet-connected systems
- Censys – Internet infrastructure and certificates
- theHarvester – Email, subdomain and host discovery
- Hunter.io – Find email addresses
- HaveIBeenPwned – Check data breaches
- Recon-ng – OSINT automation framework
- SpiderFoot – Automated OSINT collection

Tips

✓ Use multiple tools together
✓ Verify data from different sources
✓ Keep notes and organize findings
✓ Stay within legal and ethical limits

Remember

Tools help, but your skills and judgment make the difference!

Common Use Cases

Investigations → Threat Intelligence → Background Checks → Fraud Detection → Research & Analysis

Best Practices

✓ Use trusted and verified tools
✓ Do not access restricted data
✓ Respect privacy and data protection
✓ Document your process and sources
✓ Use OSINT for good and legal purposes

Ethical Reminder

- OSINT is not hacking.
- Respect laws and platform terms.
- Do not misuse personal data.
- Use for legitimate and positive purposes only.

Key Takeaway

The right tool + the right query + good verification = Powerful OSINT Results

☆ **Note** : Tools and data change frequently. Always update your knowledge and verify information before relying on it. ☺

9. Search Engines for OSINT

- Search engines are the starting point of any OSINT investigation.
- Different search engines have different strengths.

Search Engine	Best For	Key Features
Google	General information, files, web pages, dorking	• Largest index • Powerful advanced search & operators
Bing	Images, videos, academic & Microsoft resources	• Good for visual search • Shows more results than Google sometimes
DuckDuckGo	Privacy focused searches	• No tracking • Clean interface
Yandex	Russian content, deep web, files	• Excellent for Russian language • Finds files other engines miss
Baidu	Chinese web content	• Best for Chinese language • Access to Baidu specific resources
Brave Search	Privacy + relevant results	• Independent index • No profiling or tracking
Startpage	Google results with privacy	• Google powered results • Protects your identity

Tips

✓ Use multiple search engines for better results.

✓ Change language and region for more variations.

✓ Combine operators to refine searches.

✓ Always verify information from multiple sources.

Advanced Search (Google)

- Use Google Advanced Search:
  https://www.google.com/advanced_search
- You can filter by:
  → Language
  → Region
  → File type
  → Date range
  → Specific site or domain

Powerful Google Features

- Google Alerts – Get alerts on new content.
- Google Trends – See trending topics.
- Google Cache – View cached versions of pages.
- Google Images – Reverse image search.
- Google News – Find news articles.

Note

No single search engine is perfect. Use the right engine for the right job. Diversify & verify!

Example Searches (Using Google Dorks)

Purpose	Dork / Query	Description
Find PDFs	filetype:pdf cybersecurity	Find PDF files related to cybersecurity.
Find Login Pages	intitle:"login"	Find pages with login in the title.
Find Word Docs	filetype:docx "confidential report"	Find Word documents with confidential report.
Find Admin Panels	inurl:admin intitle:login	Find admin login pages.
Find Camera Feeds	inurl:view/index.shtml "live view"	Find live camera feeds.
Find Emails	intext:"@gmail.com"	Find pages containing gmail addresses.

Best Practices

✓ Be specific and targeted.
✓ Use quotation marks for exact phrases.
✓ Combine multiple operators.
✓ Review, analyze and verify results.
✓ Document your findings.

Things to Avoid

× Don't search for illegal or unethical content.
× Don't invade privacy.
× Don't rely on unverified information.
× Don't misuse OSINT for harmful purposes.

☆ A skilled investigator knows where and how to search.
Better queries → Better results → Better intelligence.

10. Social Media Intelligence (SOCMINT)

- Social media is one of the richest sources of OSINT.
- Information shared publicly (or leaked indirectly) can reveal a lot about individuals, organizations and events.

Platform	Best For	Information You Can Find
Facebook	People search, groups, events, public pages	Profiles, photos, friends, pages liked, events, posts, comments
Instagram	Images, lifestyle, locations, hashtags	Photos, stories, highlights, followers, hashtags, tagged locations
X (Twitter)	Real-time updates, conversations, trends	Tweets, replies, hashtags, trends, user info, media
LinkedIn	Professional info, companies, networking	Profiles, work history, skills, connections, company details, posts
YouTube	Videos, comments, channels	Videos, descriptions, comments, metadata, channel info
Reddit	Communities, discussions, niche topics	Posts, comments, subreddits, user activity
TikTok	Short videos, trends, creators	Videos, comments, bio, followers, trending content
Telegram	Groups, channels, leaks, discussions	Public groups/channels, files, announcements, usernames

Key Points

- ✓ Start with public information only.
- ✓ Check multiple platforms.
- ✓ Look for patterns and connections.
- ✓ Metadata, comments and likes can reveal a lot.
- ✓ People often reveal more than they realize.

Useful Tools for SOCMINT

- Maltego – Link analysis & entity relationship
- Sherlock – Find usernames across platforms
- Spokeo / PeekYou – People search engines
- HypeAuditor – Check influencer authenticity
- TweetDeck – Monitor tweets & hashtags
- Social Blade – Stats of social media accounts
- PathINT – Social media OSINT tool
- Brand24 – Monitor mentions & keywords
- Crowdtangle – Analyze public content (Meta)
- Telegram Search – Search public channels/groups

Remember

Not everything is public, and not everything public is yours to use. Respect privacy and legal limits. ☺

SOCMINT Techniques

Technique	Description
Profile Analysis	Check bio, profile picture, posts, connections, interests and activity.
Content Analysis	Analyze posts, images, videos, comments, hashtags and timing.
Network Analysis	Study friends, followers, following, groups and interactions.
Location Analysis	Look for location tags, photos, check-ins, events and local references.
Username Tracking	Search the same username across multiple platforms.
Image / Video Analysis	Use reverse image search to find source or other occurrences.
Metadata Analysis	Extract metadata from images/videos/documents (EXIF, timestamps).

Example (SOCMINT)

You find someone's public Instagram profile.
↓
Check profile & posts
↓
Note location in bio
↓
Use image search on photos
↓
Find same pics on Facebook
↓
Discover more about person.

Best Practices

- ✓ Use public information only.
- ✓ Verify information from multiple sources.
- ✓ Be patient and observe patterns.
- ✓ Keep records and screenshots (with date/time).
- ✓ Use tools ethically and responsibly.

Things to Avoid

- ✗ Hacking or unauthorized access.
- ✗ Fake accounts to gather info.
- ✗ Sharing personal data of others.
- ✗ Harassment, stalking or doxxing.
- ✗ Using OSINT for illegal activities.

Applications

- Threat intelligence & monitoring
- Background verification
- Fraud investigation
- Brand / reputation monitoring
- Missing persons / people tracing

Information is everywhere, but intelligence is in how you connect the dots.
Observe. Analyze. Verify. Respect. ☺

11. DOMAIN & NETWORK INTELLIGENCE (OSINT)

- Domain and network information can reveal ownership, infrastructure, relationships and potential risks.
- Useful for mapping digital footprint and identifying threats.

KEY POINTS

- ✓ Domains connect people, businesses and infrastructure.
- ✓ WHOIS, DNS and IP data provide valuable insights.
- ✓ Always correlate with other sources for accuracy.

A. DOMAIN INTELLIGENCE – WHAT CAN YOU FIND?

INFORMATION	DESCRIPTION	EXAMPLES / USE CASES
Domain Registration (WHOIS)	Details about domain owner, registrar, registration & expiration dates, nameservers.	Identify owner, registration dates, changes in ownership.
DNS Records	Records that map domain to IP and define email, subdomains, services.	A, AAAA, MX, NS, TXT, CNAME, SPF records.
IP Address Information	IP address, ISP, organization, location, ASN details.	Find hosting provider, ISP, organizational owner.
Subdomains	Subdomains can expose internal services or applications.	blog.example.com, mail.example.com, portal.example.com
Hosting & Infrastructure	Web server, CDN, technologies and hosting history.	Detect stack, CDN usage, server technologies.
Domain Relationships	Connected domains, common registrant, aliases.	Find related websites, sister companies or fake domains.

REMEMBER

WHOIS data may be hidden by privacy protection services.

Use multiple tools and sources to confirm information.

Data changes over time – always verify.

COMMON USE CASES

- Threat intelligence & attribution
- Fraud & phishing investigation
- Brand protection
- Attack surface discovery
- Competitor analysis.

B. POPULAR TOOLS FOR DOMAIN & NETWORK OSINT

WHOIS	DNSDumpster	MXToolbox	SecurityTrails	Shodan	Censys	ViewDNS.info
Domain registration and ownership details.	DNS lookup and reconnaissance tool.	Check DNS, MX, Blacklist, SPF & more.	Historical DNS, Subdomains, IP history.	Search for devices and services on the internet.	Internet asset search engine.	DNS lookup & propagation tools.

C. COMMON DNS RECORDS – QUICK REFERENCE

RECORD TYPE	PURPOSE	EXAMPLE
A	Maps domain to IPv4 address	example.com → 93.184.216.34
AAAA	Maps domain to IPv6 address	example.com → 2001:db8::1
MX	Mail Exchange record	example.com → mail.example.com
NS	Name Server record	example.com → ns1.example.com
TXT	Text record (SPF, verification, etc.)	v=spf1 include:_spf.example.com ~all
CNAME	Alias of another domain	www.example.com → example.com
SOA	Start of Authority record	Primary DNS server information
PTR	Reverse DNS (IP to domain)	93.184.216.34 → example.com

D. EXAMPLE WORKFLOW (DOMAIN OSINT)

1. **Identify Target Domain**
   Start with domain name.
2. **WHOIS Lookup**
   Find registration and ownership details.
3. **DNS Enumeration**
   Check DNS records, subdomains & email configuration.
4. **IP & Infrastructure Lookup**
   Find IP, ISP, ASN, location, hosting info.
5. **Correlate & Analyze**
   Correlate with other OSINT data for full picture.

E. BEST PRACTICES

- ✓ Always verify information from multiple sources.
- ✓ Look for historical data (changes over time).
- ✓ Map relationships between domains, IPs and ASNs.
- ✓ Document findings with date, source and methodology.
- ✓ Use OSINT ethically and for lawful purposes only.

F. THINGS TO AVOID

- ✗ Relying on a single source of data.
- ✗ Assuming WHOIS info is always accurate.
- ✗ Ignoring historical or passive DNS data.
- ✗ Scanning or probing without authorization.
- ✗ Using OSINT for illegal or unethical activities.

"Every domain leaves footprints. Your job is to find them, connect them, and understand the story."

Find • Correlate • Analyze • Verify

12. GEOLOCATION & MAPPING (OSINT)

Geolocation helps identify the real-world location of people, objects or events using digital clues.

Useful for investigating images, tracking activities, analyzing incidents, and understanding physical context.

KEY POINTS

- ✓ Visual clues can reveal powerful location information.
- ✓ Small details (shadows, signs, plate numbers, architecture) matter.
- ✓ Always cross-check using multiple sources.
- ✓ Accuracy improves with practice and tools.

A. WHAT CAN YOU FIND USING GEOLOCATION?

CATEGORY	WHAT YOU CAN FIND	EXAMPLES / USE CASES
Exact Location	Coordinates (lat/long), addresses, places.	Find where a photo was taken, locate a business, identify an area.
Physical Features	Buildings, roads, landmarks, terrain, nature.	Match background features in images or videos.
Time & Weather	Local time, timezone, weather conditions.	Verify time of event, check weather at that location.
Satellite View	High-resolution satellite imagery, changes over time.	Analyze infrastructure, land use, and environment.
Nearby Places	Nearby businesses, streets, points of interest.	Understand surroundings and confirm public activity.
Network Location	Approximate location from IP, Wi-Fi or cell data.	Track online activity origin, verify claims.
Historical Data	Past imagery, map changes, older views.	See how a place changed over time.

B. POPULAR GEOLOCATION TOOLS

Google Maps	Explore places, streets, routes, and business info.
Google Earth	3D view, historical imagery, measure distances.
GeoGuessr	Game-based tool to guess locations from street view.
Wikimapia	User-edited map with details and photos.
OpenStreetMap	Open-source map data for everything.
Sentinel Hub	Access satellite imagery (European Space Agency).
ExifTool	Extract metadata (GPS coordinates) from files.
IPinfo / ipinfo.io	Find approximate location from IP address.

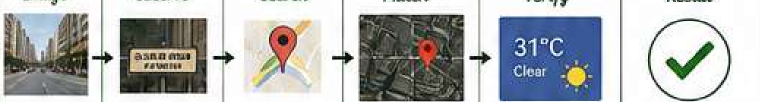
C. GEOLOCATION TECHNIQUES (COMMON METHODS)

1	Image Clues	Analyze buildings, street signs, billboards, vegetation, mountains, vehicles, license plates.
2	Shadow Analysis	Use shadow direction & length with SunCalc.org to estimate time and location.
3	Text & Language	Check language, scripts, phone numbers, URLs, advertisements, shop names.
4	Weather & Environment	Match weather, season, and natural conditions with local climate data.
5	Infrastructure & Layout	Study road patterns, bridges, building styles, traffic signs, and public transport.
6	Satellite Imagery	Use high-res satellite views to match locations and confirm details.
7	Digital Signals	Use Wi-Fi SSID names, cell towers or IP data to narrow down location.

D. GEOLOCATION WORKFLOW (EXAMPLE)

1. **Collect**
   Get the image, video, post or digital clue.
2. **Observe**
   List all visible clues and details.
3. **Analyze**
   Use techniques (shadows, text, landmarks, architecture, environment).
4. **Search**
   Use tools (Maps, Earth, GeoGuessr, Satellite) to find possible locations.
5. **Verify**
   Cross-check with multiple sources and confirm accuracy.
6. **Document**
   Record findings, sources, screenshots & confidence level.

E. EXAMPLE – HOW A LOCATION CAN BE FOUND



1 Image	2 Observe	3 Search	4 Match	5 Verify	6 Result
Photo with buildings, palm trees & road sign.	Arabic text on sign, yellow license plate, specific architecture.	Search similar buildings & road layout on Maps.	Match with Street View & satellite imagery.	Check weather & time to confirm location/time.	Location found with high confidence.

E. BEST PRACTICES

- ✓ Use multiple tools and verify results.
- ✓ Look at small details; nothing is useless.
- ✓ Compare with different dates and angles.
- ✓ Keep a record of sources and screenshots.
- ✓ Respect privacy and legal boundaries.

G. THINGS TO AVOID

- ✗ Jumping to conclusions without confirmation.
- ✗ Relying on a single clue or tool.
- ✗ Sharing exact locations of private individuals.
- ✗ Using geolocation for stalking or harassment.
- ✗ Ignoring time differences and seasonal changes.

H. APPLICATIONS

- Threat intelligence & investigations
- Missing persons / disaster response
- Journalism & fact-checking
- Fraud & cybercrime investigations
- Military, defense & risk assessment

Every place leaves clues. Good eyes see more, tools verify it, and ethics guide our path.

Observe · Correlate · Verify · Respect

13. DATA & DOCUMENT ANALYSIS (OSINT)

Documents, files and public datasets contain valuable information.

Metadata and hidden details can reveal more than the content itself.

KEY POINTS

- ✓ Metadata reveals origin, device, time and modification details.
- ✓ Hidden data can expose systems, people and locations.
- ✓ Always verify authenticity and cross-check multiple sources.
- ✓ Data analysis converts raw information into intelligence.

A. WHAT CAN YOU FIND?

TYPE	WHAT YOU CAN FIND	EXAMPLES / USE CASES
Documents (PDF, DOCX, PPTX)	Text content, author, company, creation date, software used, comments, hidden metadata.	Leaked reports, proposals, invoices, policies, contracts.
Images (JPG, PNG, etc.)	EXIF data, GPS coordinates, device info, timestamps, editing history.	Location, camera model, capture time, orientation.
Videos	Metadata, codecs, resolution, duration, frames, thumbnails, location (if available).	Event verification, location clues, timing & sequence.
Archives (ZIP, RAR, 7z)	File list, paths, hidden files, creation date, comments.	Extract sensitive files, recover deleted items.
Emails (EML, MSG)	Headers, sender/receiver, IP addresses, timestamps, route.	Phishing analysis, origin tracking, timeline building.
Datasets (CSV, XLSX, JSON)	Rows, columns, patterns, IDs, relationships, outliers, trends and statistics.	Public data analysis, fraud detection, business intelligence.
Code & Config Files	APIs, keys, credentials, paths, comments, user info.	Exposed secrets, misconfigurations, internal structure.

B. POPULAR TOOLS

ExifTool	Extract metadata from every file type.
M2G Metadata2Go	View metadata of images, documents, videos.
FOCA	Extract metadata & find hidden info in docs.
Doc Analyzer (Microsoft)	Analyze Office docs for hidden data.
PDFid / PDFINFO	Extract object info from PDF files.
Wireshark	Analyze network captures and file transfers.
Strings (Linux)	Find readable strings in files and binaries.
CyberChef	Perform data extraction & decoding.

C. METADATA – WHAT IT CAN REVEAL

- Author / Creator
- Creation Date / Time
- Last Modified
- Software / App Used
- Device / Camera Info
- GPS Coordinates
- File Path / Location
- Comments / Revisions

Metadata is often hidden but always valuable.
Always inspect before you trust.

D. DATA & DOCUMENT ANALYSIS TECHNIQUES

1	Metadata Extraction	Extract and review metadata using tools (ExifTool, Metadata2Go, etc.).
2	Content Review	Read and analyze visible content for key information and context.
3	Hidden Data Discovery	Find hidden objects, comments, tracked changes, embedded files.
4	Data Validation	Verify dates, names, numbers and facts with other reliable sources.
5	Correlation	Connect information with other OSINT data to build stronger intelligence.
6	Pattern & Trend Analysis	Identify patterns, anomalies and trends in datasets and documents.
7	Reporting	Document findings clearly with evidence, screenshots and source references.

E. EXAMPLE – DOCUMENT OSINT WORKFLOW

1. **Collect** → Obtain the document/file from public or leaked source.
2. **Extract Metadata** → Use ExifTool or other tools to extract metadata.
3. **Analyze Content** → Read content, check authors, dates, keywords.
4. **Cross-Check** → Verify info with other sources (OSINT tools).
5. **Correlate & Interpret** → Connect dots, find relations and meaning.
6. **Report Findings** → Prepare report with evidence and confidence.

F. COMMON FILE TYPES & WHAT TO CHECK

- PDF – Metadata, authors, software, hidden objects
- DOCX – Document properties, comments, revisions
- XLSX/CSV – Hidden sheets, formulas, data patterns
- JPG/PNG – EXIF data, GPS, device, timestamps
- MP4/AVI – Metadata, encoding, location (if any)
- ZIP/RAR – File list, paths, hidden files
- EML/MSG – Headers, IPs, routes, timestamps

G. BEST PRACTICES

- ✓ Always verify with multiple sources.
- ✓ Check metadata even if content looks normal.
- ✓ Keep original files for re-verification.
- ✓ Document every step and tool you use.
- ✓ Respect privacy and legal boundaries.

H. THINGS TO AVOID

- ✗ Don't trust a single source or file.
- ✗ Don't share sensitive or personal data.
- ✗ Don't modify original evidence files.
- ✗ Don't ignore timestamps or context.
- ✗ Don't use OSINT for unethical or illegal purposes.

I. APPLICATIONS

- Threat intelligence & attacker profiling
- Fraud investigation & digital forensics
- Leak analysis & data breach investigation
- Business intelligence & competitor research
- Fact-checking & event verification

Every file tells a story. Extract. Analyze. Correlate. Verify.
Information + Context + Verification = Intelligence

14. PUBLIC RECORDS & OPEN DATA (OSINT)

Public records and open data are reliable sources of OSINT.

Government, legal and public databases provide factual and verifiable information about people, organizations and assets.

KEY POINTS

- ✓ Public records are generally reliable and legally accessible.
- ✓ Information may be scattered across multiple portals and formats.
- ✓ Always cross-check and verify data from multiple sources.
- ✓ Open data enables transparency and better decision-making.

A. WHAT CAN YOU FIND?

CATEGORY	WHAT YOU CAN FIND	EXAMPLES / USE CASES
Identity Records	Names, aliases, DOB, addresses, relatives, nationality, ID numbers.	Background checks, identity verification, connections.
Court & Legal Records	Lawsuits, case details, judgments, charges, filings.	Litigation research, fraud checks, asset disputes.
Business & Company Data	Company registration, directors, ownership, filings, financials.	Due diligence, competitor research, KYC.
Property & Land Records	Property ownership, deeds, mortgages, land parcels, value.	Ownership verification, real estate research.
Vehicle Records	Registration, owner details, insurance status, accidents.	Ownership lookup, history checks, investigations.
Voter & Electoral Data	Voter lists, electoral rolls, ward/booth information.	Demographic analysis, location intelligence.
Financial & Tax Data	Tax liens, assessments, budgets, public spending.	Financial profiling, risk assessment.
Licenses & Permits	Professional licenses, permits, registrations, certifications.	Verify legitimacy, compliance and credentials.
Open Government Data	Census, crime data, budgets, health, education, transport, environment, more.	Trend analysis, mapping, reports, research.

B. POPULAR SOURCES & PORTALS

USA.gov	U.S. government services and public information.
data.gov	Open data portal (US) – datasets & reports.
CourtListener	U.S. federal court records and cases.
OpenCorporates	Company data from across the world.
PropertyShark	Real estate & property records (US).
VehicleHistory	Vehicle history and ownership data.
State / Local Portals	State or local government records and databases.
OpenDataSoft	Global open data directory.
Kaggle Datasets	Datasets for research and analysis.

C. TYPES OF PUBLIC RECORDS (EXAMPLES)

Birth / Death Records	Marriage / Divorce Records	Criminal Records (If Public)	Professional License Records
UCC Filings (Business Liens)	Tax Assessor Records	Utility Records (If Public)	School & Education Records
Business Registry Records	Environmental Permits	Campaign Finance Records	Public Contracts & Procurement

D. HOW TO FIND PUBLIC RECORDS (WORKFLOW)

1. **Identify What You Need** — Define the person, business, asset or event.
2. **Find Relevant Sources** — Search government portals, databases & directories.
3. **Access & Download Data** — Use search, filters or requests to obtain records.
4. **Analyze & Cross-Check** — Verify accuracy and compare across sources.
5. **Correlate & Build Intelligence** — Connect dots and extract meaningful insights.
6. **Document & Report** — Record findings with source references.

E. EXAMPLE – PUBLIC RECORDS OSINT WORKFLOW

1. Define Target	2. Search Sources	3. Retrieve Data	4. Verify Data	5. Analyze	6. Report
Identify the person/company or asset.	Find relevant public portals and databases.	Download or request the records.	Cross-check with multiple reliable sources.	Extract key information and identify links.	Document findings with citations and evidence.

F. COMMON SEARCH STRATEGIES

- Use official portals first.
- Try variations of name, business, address.
- Use filters: date range, location, document type.
- Check state, county, city level records.
- Search by IDs: tax ID, license, case number, VIN.
- Combine with other OSINT sources for context.

G. BEST PRACTICES

- ✓ Use official and trusted portals.
- ✓ Verify data from multiple sources.
- ✓ Note the source, date and URL.
- ✓ Keep records organized and updated.
- ✓ Respect privacy and legal boundaries.

H. THINGS TO AVOID

- ✗ Do not use fake or misleading data.
- ✗ Do not assume data is 100% accurate.
- ✗ Avoid old records without verifying recency.
- ✗ Do not access data that requires authorization.
- ✗ Do not use OSINT for harassment, stalking or illegal activities.

I. APPLICATIONS

- Background checks & due diligence
- Fraud detection & investigation
- Threat intelligence & risk assessment
- Legal research & case preparation
- Business intelligence & market research
- Journalism & fact-checking

"Public records are public for a reason — transparency empowers everyone."

Find • Verify • Analyze • Correlate • Report

Reliable data → Accurate insights → Better decisions

OSINT & GOOGLE DORKING

1. Introduction to OSINT

- OSINT stands for Open-Source Intelligence.
- It refers to collecting and analyzing information from publicly available sources.
- The goal of OSINT is to turn openly available data into useful intelligence.
- It plays a vital role in cybersecurity, threat intelligence, fraud investigation, risk assessment and more.

OSINT = DATA → INFORMATION → INTELLIGENCE

2. Why OSINT is Important?

- Helps in understanding threats and attackers.
- Identifies exposed or leaked information.
- Supports security assessments and audits.
- Useful for investigation and incident response.
- Helps organizations make informed decisions.

Benefits
- Cost effective
- Legal and ethical (when done properly)
- Wide range of sources
- Real-time information

3. Types of OSINT Sources

- **Public Web**: Websites, Blogs, News sites, Forums, Online Documents
- **Social Media**: Platforms like Facebook, Twitter (X), LinkedIn, Instagram etc.
- **Public Records**: Govt records, Court documents, Business registries, Voter lists, etc.
- **Technical Sources**: Domain info (WHOIS), DNS records, IP info, Shodan, Censys, Certificates
- **Media & Archives**: News archives, Wayback Machine, YouTube, Podcasts, Press releases

4. OSINT Lifecycle

- **Planning** → Define objective, scope, target and legal boundaries.
- **Collection** → Gather data from various public sources.
- **Processing** → Organize, filter and clean the collected data.
- **Analysis** → Analyze the data to find patterns, connections and insights.
- **Reporting** → Document findings in a clear and structured format.

5. OSINT Collection Techniques

- Direct Search – Using search engines
- Social Media Profiling – Analyzing user information, posts, followers, likes, etc.
- Website Footprinting – Extracting details about websites.
- Metadata Analysis – Checking hidden information in files and images.
- Public Records Search – Using government and public databases.
- Archive Search – Using Wayback Machine and other archives.
- Domain & IP Recon – WHOIS, DNS, IP lookup, reverse IP lookup.

6. Google Dorking – Introduction

- Google Dorking is the practice of using advanced search operators to find specific information on the internet.
- It helps in finding hidden pages, sensitive files, login portals, leaked data, misconfigurations and more.
- Dorking is not hacking. It uses Google's search engine in a smart way.

Important
Use Google Dorking only on systems and domains you are authorized to test.

7. How Google Dorking Works?

- Google uses special operators to filter search results.
- These operators help us search in titles, URLs, file types, websites and content.
- By combining operators, we can find highly specific results.

Google → Operators (site:, filetype:, intitle:, inurl:, " ", OR, -) → Refined Results

8. Common Google Dorks (Operators)

Operator	Description	Example
site:	Search within a specific website	site:example.com
filetype:	Search for specific file types	filetype:pdf
intitle:	Search keyword in title	intitle:"login"
inurl:	Search keyword in URL	inurl:admin
" "	Search exact phrase	"privacy policy"
OR	Search one term or another	login OR sign in
–	Exclude a term	password -admin

9. Google Dorks – Examples (Safe)

- site:example.com → All pages from example.com
- site:example.com filetype:pdf → PDF files from example.com
- site:example.com intitle:"privacy policy" → Pages with privacy policy in title
- site:example.com inurl:login → URLs containing login
- site:example.com "terms and conditions" → Pages with exact phrase
- site:example.com filetype:xls OR filetype:xlsx → Excel files in example.com

10. Advanced Google Dorking Examples

- site:edu filetype:pdf "course syllabus"
- inurl:admin filetype:php
- site:gov inurl:contact
- intitle:"index of" "parent directory"
- filetype:log "error"
- site:example.com -inurl:example2.com
- "confidential" filetype:docx

Note: Always ensure you have permission and are following legal and ethical guidelines.

11. OSINT Tools – Categories

- Search Engines → Google, Bing, DuckDuckGo
- Whois Lookup → whois.domaintools.com
- DNS Lookup → dnsdumpster.com, mxtoolbox.com
- IP & Network → ipinfo.io, Shodan.io, censys.io
- Social Media → Maltego, SpiderFoot, Sherlock
- Archive & Cache → Wayback Machine, Google Cache
- Metadata Tools → exiftool, metapicz, online viewers
- Reporting → Google Docs, Sheets, Markdown

12. WHOIS Lookup

- WHOIS provides information about domain registration.
- We can find:
  - ✓ Domain registrar
  - ✓ Registrant name
  - ✓ Contact email
  - ✓ Creation date
  - ✓ Expiration date
  - ✓ Name servers

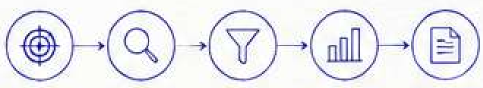
Domain Name: example.com
Registrar: Example Registrar
Registrant: John Doe
Email: john@example.com
Created On: 2021-01-01
Expires On: 2026-01-01
Name Servers:
ns1.example.com
ns2.example.com

13. DNS & IP Recon

- DNS lookup helps to find records related to a domain.
- Common DNS records:
  - A Record → Maps domain to IP
  - MX Record → Mail server
  - NS Record → Name servers
  - TXT Record → Text information
  - CNAME Record → Canonical name

Tools: dnsdumpster.com, mxtoolbox.com

14. OSINT Investigation Flow



1. Define Objective – What do you want to find?
2. Collect Data – Gather data from public sources.
3. Verify & Clean – Remove duplicates and verify accuracy.
4. Analyze – Find patterns, connections and insights.
5. Report – Create clear and structured report.

15. Ethical Guidelines

→ Use only publicly available information.
→ Do not access private or restricted systems.
→ Do not attempt to exploit vulnerabilities.
→ Respect privacy and data protection laws.
→ Use information only for authorized purposes.
→ Always give credit to original sources.
→ Verify information before reporting.

Be Ethical, Be Legal, Be Responsible.

16. Common Mistakes to Avoid

✗ Using OSINT for illegal or unethical activities.
✗ Relying on single source (verify always).
✗ Ignoring data freshness (old data can be misleading).
✗ Not documenting sources.
✗ Exposing sensitive information accidentally.

Good OSINT = Right information from Right source at Right time.

17. Practical OSINT Workflow (Example)

- Target → example.com
- Search Engine → site:example.com filetype:pdf
- Findings → policy.pdf, report.pdf, data.xlsx
- Social Media → LinkedIn, Twitter posts about company
- Whois & DNS → Get domain, registrar and IP info
- Archive → Check old websites on Wayback
- Analysis → Correlate all data and find insights
- Report → Prepare and share final report

18. Conclusion

- OSINT is a powerful skill in cybersecurity and investigations.
- It helps in gathering valuable intelligence from open and public sources.
- Google Dorking is an important part of OSINT which helps to find hidden and specific information.
- When used ethically and responsibly, OSINT can greatly improve security awareness and decision making.

Information is everywhere.
Intelligence is what you make of it.